

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: COMMERCIAL ARITHMETIC 1 (SUB-STRAND 2.8) FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Amina is a Form 4 student from Kisumu. She saved KSh 4,500 from selling mandazi at school. She wants to use it to (1) contribute to the Environmental Club's end-of-term trip to Lake Victoria, (2) buy a second-hand smartphone at a Kondele electronics stall advertised at 20% off, and (3) send the remaining money to her aunt in Uganda through a mobile-money agent. Use what you learned in Lessons 1 to 8 to answer the questions below. Write your answers in the space under each question. Show every step of your working, write the formula you use, and include units (KSh or UGX). Explain your reasoning in full sentences where the question asks you to explain. Your final answer should show how commercial arithmetic helps Amina make smart, fair and informed money decisions.

Part 1: Why Amina needs a budget

The Environmental Club plans a one-day trip to Lake Victoria. Its total expenditure is KSh 12,250 and its income is KSh 6,000. (a) Calculate the difference between the club's income and expenditure, and state what this difference is called. (b) The cost list includes bus hire and entry fees. Say which is a fixed cost and which is a variable cost, and explain how you can tell. (c) Amina has not yet written down her own costs. Using the club's situation as evidence, explain why she should prepare a budget before she starts spending her KSh 4,500.

(a) Deficit = Expenditure – Income = 12,250 – 6,000 = KSh 6,250. Because expenditure is greater than income, this shortfall is called a deficit. (b) Bus hire is a fixed cost: the bus costs the same whether 10 or 30 students travel. Entry fees are a variable cost: they increase with every extra participant, because each person pays. (c) A budget is a plan that compares income (money coming in) with expenditure (money going out). The club only discovered its KSh 6,250 deficit after listing and totalling all its costs. If Amina does not list her three planned expenses and compare them to her KSh 4,500, she cannot know whether she has enough or how much she is short until it is too late. A budget lets her see the problem early and decide how to solve it. That is why we need a budget.

Part 2: Discounts and which deal is better

Amina first gives KSh 800 to the Environmental Club, which

(a) Discount = $20/100 \times 3,200$ = KSh 640. Sale price = 3,200 – 640 = KSh 2,560. Amina has KSh 3,700, so she can afford it. Money left = 3,700 – 2,560 = KSh 1,140. (b) Discount = $25/100 \times 3,440$ = KSh 860. Sale price = 3,440 – 860 = KSh 2,580. (c)

leaves her with KSh 3,700. (a) The Kondele stall advertises the phone at a marked price of KSh 3,200 with 20% off. Calculate the discount and the sale price, and say whether Amina can afford it from her KSh 3,700. How much would she have left? (b) A Naivas supermarket advertises a similar phone at a marked price of KSh 3,440 with 25% off. Calculate its sale price. (c) Amina says: 'Naivas has the bigger percentage discount, so it must be the better deal.' Use your answers to decide whether she is correct, and explain how to compare discount offers fairly. (d) Explain why, in your calculation of the 20% discount, you were checking the advertisement with mathematics rather than just trusting the sign.	Amina is not correct. Kondele costs KSh 2,560 and Naivas costs KSh 2,580, so Kondele is KSh 20 cheaper even though its percentage discount is smaller. A percentage is taken from a different marked price in each shop, so percentages alone cannot be compared. To compare fairly, calculate the actual sale price (Sale Price = Marked Price – Discount) for each offer and compare the amounts in shillings. (d) A sign saying '20% off' only makes a claim. Calculating Discount = Marked Price – Sale Price, or Percentage Discount = (Discount ÷ Marked Price) × 100, shows whether the saving is real and what Amina actually pays. This is how she makes an informed decision. Here the 20% discount is genuine and gives her the cheaper phone.
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Part 3: Commission and crossing the border

Amina has KSh 1,140 left to send to her aunt in Kampala. The mobile-money agent at Kondele charges a commission of 3% of the amount sent. (a) Calculate the commission the agent earns and the amount that is left to be converted. (b) The bureau's buying rate is 27.80 UGX per KES. Calculate how many Uganda Shillings her aunt receives, to the nearest shilling. (c) Amina saw a rate of 28.50 UGX per KES yesterday and expected to use that. Explain whether it is the right rate to use for her transaction and why. (d) Explain how the forex bureau's buying and	(a) Commission = $\frac{3}{100} \times 1,140 = \text{KSh } 34.20$. Amount left = $1,140 - 34.20 = \text{KSh } 1,105.80$. (b) UGX = $1,105.80 \times 27.80 = \text{UGX } 30,741.24$, which is about UGX 30,741. (c) No. Amina is giving the bureau Kenya Shillings, so the bureau is buying her KES and uses its buying rate of 27.80 UGX per KES, which is less favourable to her. The 28.50 rate is not the rate she transacts at, and using it would overestimate what her aunt receives. The direction of the transaction (KES to UGX) decides which rate applies. (d) A bureau buys and sells currency at two different rates, and the gap between them (the spread) is its profit. This works like a percentage commission, but it is hidden inside the rate rather than charged as a separate fee. The mobile-money agent's 3% commission is visible and is deducted openly. Both reduce the value Amina's aunt actually receives, so Amina must check both.
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selling rates can work like a hidden commission, and compare this with the agent's visible commission.	
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Part 4: Profit, loss and fair prices

Every price Amina meets in the market has a profit margin built into it. Suppose the Kondele stall paid KSh 2,000 for the phone and sold it to Amina at the discounted price you found in Part 2. (a) Calculate the stall's profit. (b) Calculate the percentage profit, and state which amount the percentage is calculated on. (c) Does the stall make a loss by offering the 20% discount? Explain what this tells Amina about whether the discount is fair to both her and the trader. (d) Amina's own mandazi business also has costs. Explain why she should use her total cost price, including overheads such as fuel, when she works out her percentage profit.	(a) Selling price is KSh 2,560 and cost price is KSh 2,000. Profit = SP – CP = 2,560 – 2,000 = KSh 560. (b) Percentage Profit = $(\text{Profit} \div \text{CP}) \times 100 = (560 \div 2,000) \times 100 = 28\%$. The percentage is always calculated on the cost price. (c) No. The selling price is higher than the cost price, so the stall still makes a 28% profit even after the 20% discount. The deal is therefore fair to both sides: Amina saves KSh 640 off the marked price, and the trader still earns a profit and stays in business. (d) Percentage profit must be based on the total cost price, which is the direct costs plus overheads. If Amina leaves out overheads such as charcoal or fuel, her cost looks smaller and her profit looks bigger than it really is. Using total cost price gives an honest picture of whether her mandazi business really makes money.
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Part 5: Amina's complete financial plan and the driving question

Bring all your work together. (a) Write Amina's complete plan as a short budget starting from her KSh 4,500. Show the three allocations (the club contribution, the phone, the Uganda transfer), the commission, and what her aunt receives in UGX. (b) The same KSh 4,500 seems to produce different outcomes depending on discounts, commissions,	(a) Income: KSh 4,500. Expenditure: club contribution KSh 800; smartphone at 20% off KSh 3,200 = KSh 2,560; amount for the Uganda transfer = 4,500 – 800 – 2,560 = KSh 1,140. Commission at 3% = KSh 34.20, leaving KSh 1,105.80. Converted at 27.80 UGX per KES, her aunt receives about UGX 30,741. The total spent is 800 + 2,560 + 1,140 = KSh 4,500, so there is no deficit. (b) The discount cuts the phone price by KSh 640, which is what makes the plan affordable. The commission removes KSh 34.20 from the transfer. The exchange rate decides how many UGX the remaining money becomes, and the rate used depends on the direction of the transaction. Profit margins are built into every price, as with the stall's 28% profit, so prices that look generous still pay the trader. Each operation applies a different formula to a different base amount, so the outcomes differ even though the starting money is the same. (c) We need a budget to compare income and expenditure before spending, so that we spot a deficit early and plan how to close it. Discounts are necessary because they attract buyers and let traders sell stock, while buyers save money. Commissions are necessary because agents and intermediaries provide a service that makes trade possible, such as moving money, and they must be paid for it. (d) Amina can plan with a budget, verify discounts by calculating the sale price, compare offers using actual shillings, deduct commissions, use the correct exchange rate for the direction of the transaction, and
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exchange rates and profit or loss. Explain how each of these changes what Amina ends up with. (c) Answer the two key inquiry questions in your own words: Why do we need a budget? Why are discounts and commissions necessary in trade? (d) Answer the driving question: How can Amina, and every Kenyan, use commercial arithmetic to make smart, fair and informed money decisions in markets, clubs and across borders?	check profit or loss as a percentage of total cost. By applying these formulas in the right order to the right base values, every Kenyan can make decisions that are smart, fair and informed.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accuracy of calculations	All calculations (deficit, discount, sale price, commission, UGX conversion, profit and percentage profit) are correct, use the right formulas and the right base amounts, and carry through correctly from one step to the next.	Most calculations are correct and use the right formulas, with one or two minor arithmetic or rounding slips that do not change the overall plan.	Several calculations are incorrect or formulas are missing or misused, for example using the wrong base for a percentage or the wrong exchange rate.
Use of concepts and terms	Uses terms such as fixed and variable cost, deficit, marked price, sale price, commission, buying rate, selling rate, spread, cost price and percentage profit correctly and precisely throughout.	Uses most terms correctly, with occasional imprecision.	Terms are missing, confused or used incorrectly.
Reasoning and explanation	Explains clearly why percentages cannot be compared alone, why the exchange rate depends on direction, why the spread is a hidden commission, and why percentage profit uses cost price, with evidence from the calculations.	Gives mostly correct explanations, but some lack evidence or full reasoning.	Explanations are missing, vague or incorrect, or are not supported by the calculations.
Integration of the whole plan	Presents a complete, consistent plan in which each step feeds into the next (budget, discount, commission, conversion) and accounts for all KSh 4,500, and links it clearly to the driving question.	Presents a mostly complete plan with the main steps linked, but minor gaps in the connections or the link to the driving question.	The plan is incomplete or the steps are not linked, and little connection is made to the driving question.

Kenyan context and presentation	Keeps Amina's Kenyan context central, shows working clearly with units and steps, and writes in an organised, easy-to-follow way.	Working is mostly clear with units, and the context is mostly used.	Working is hard to follow, units are often missing, and the context is not used.
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